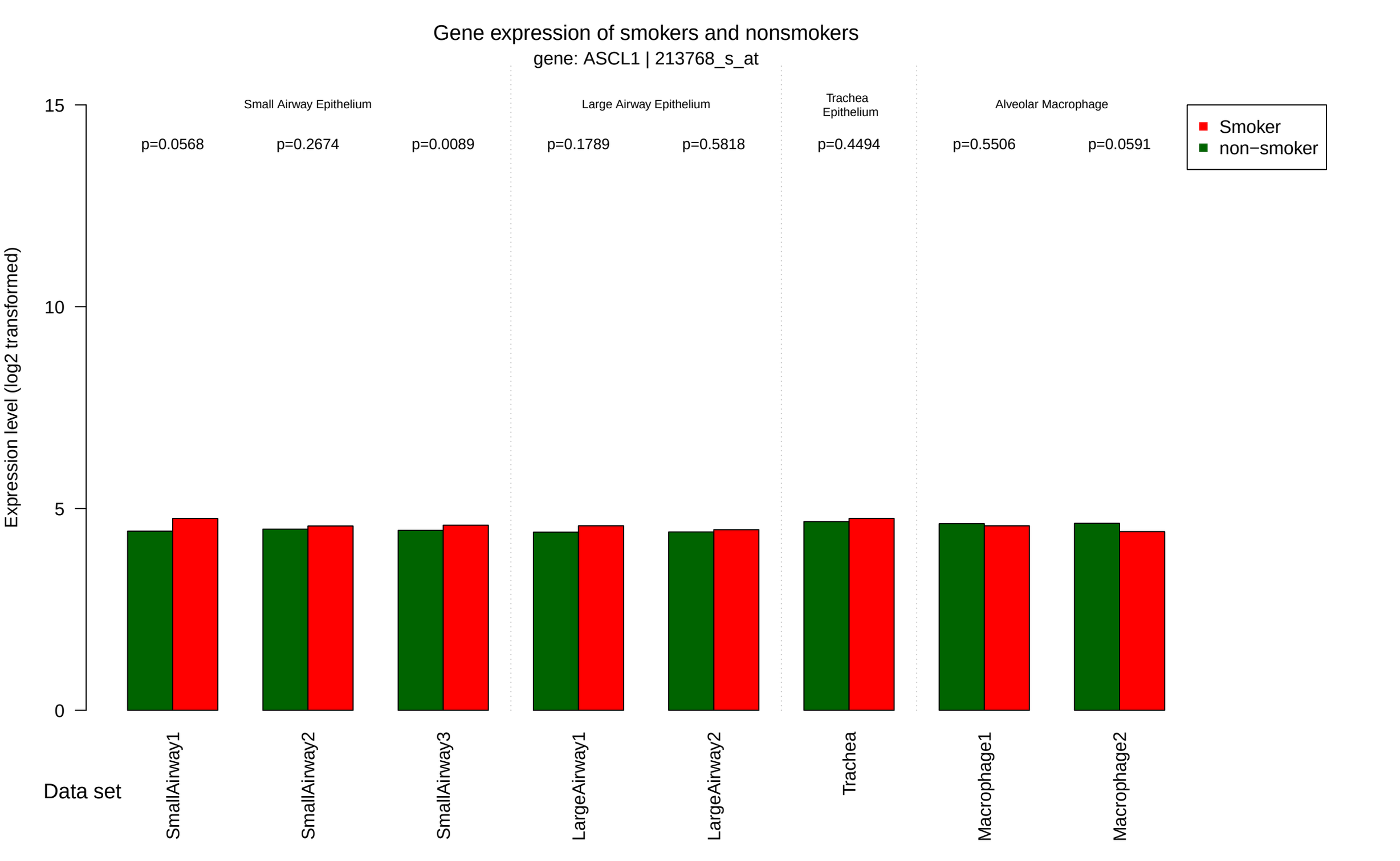
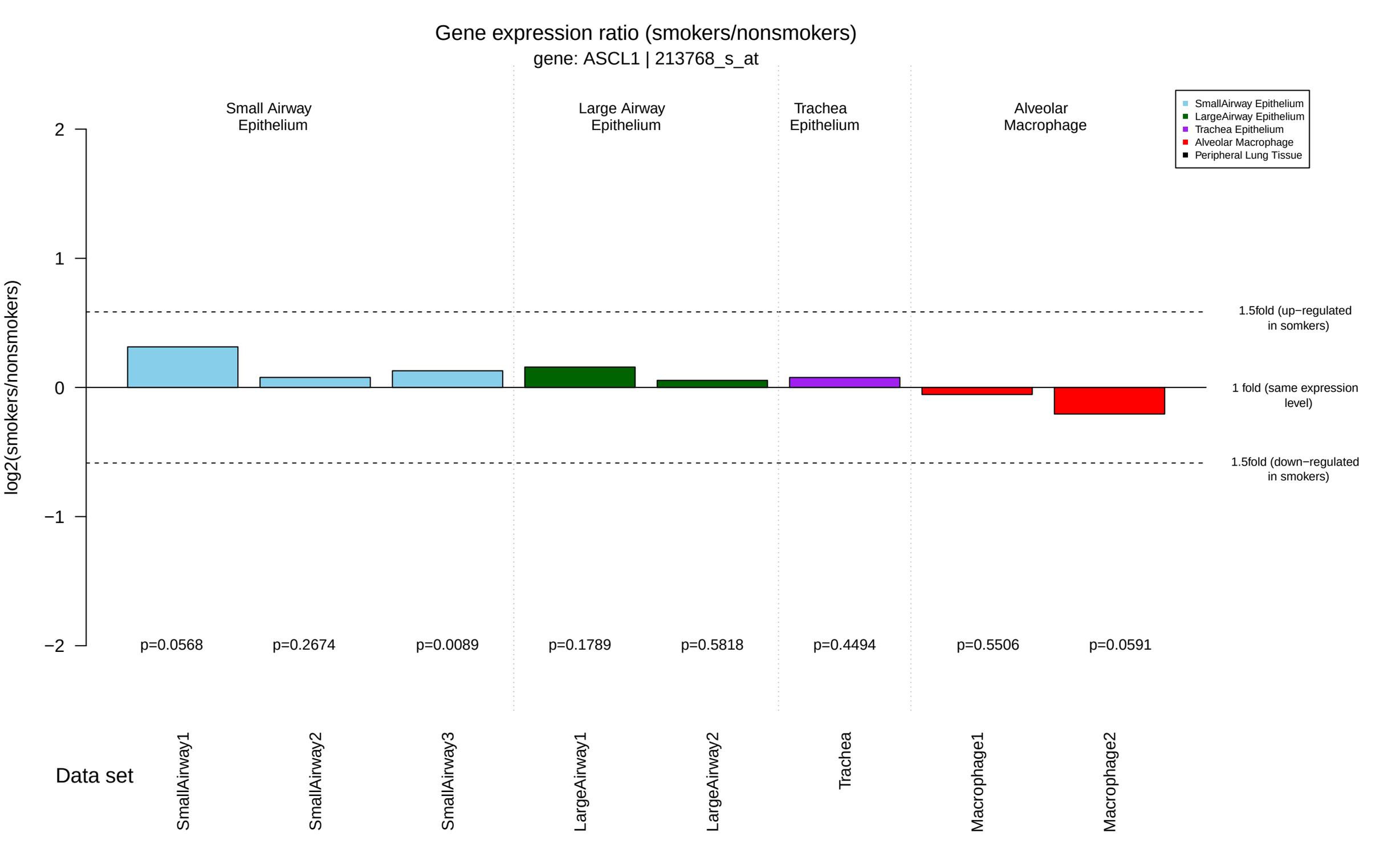
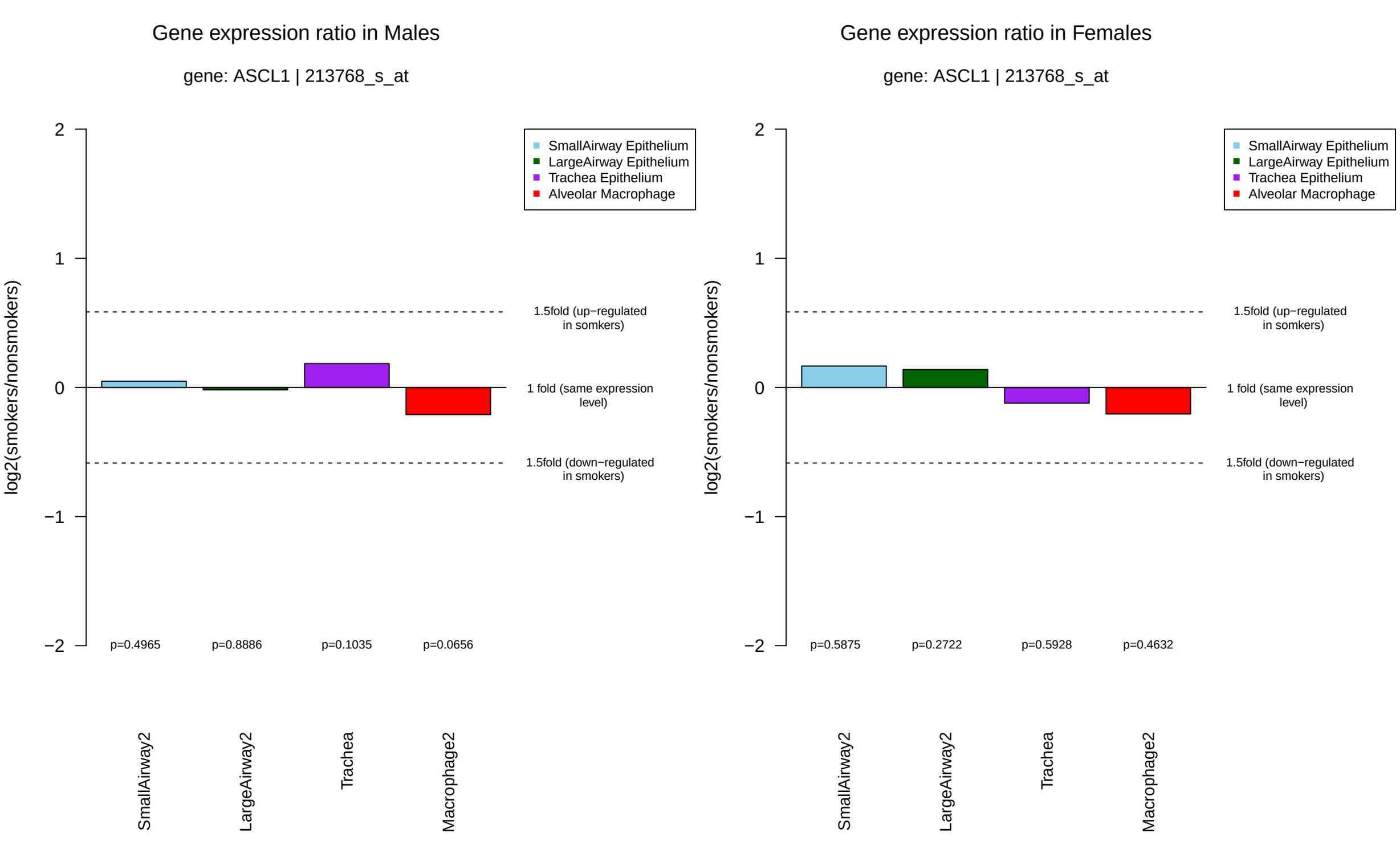
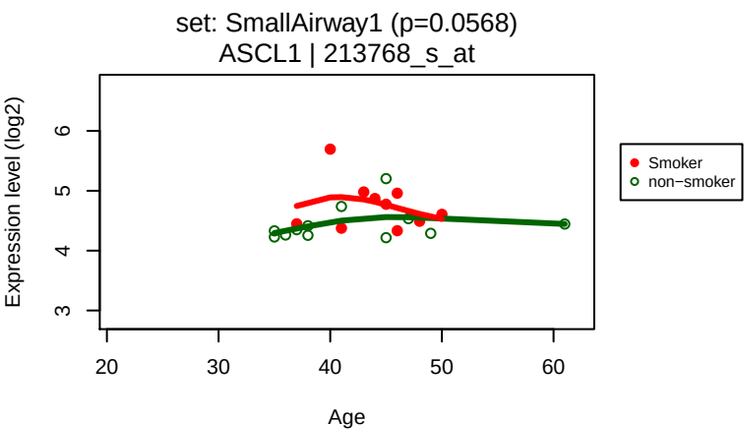
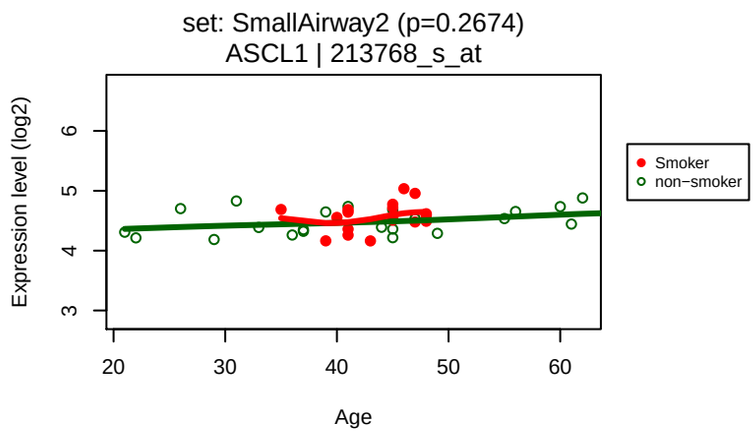
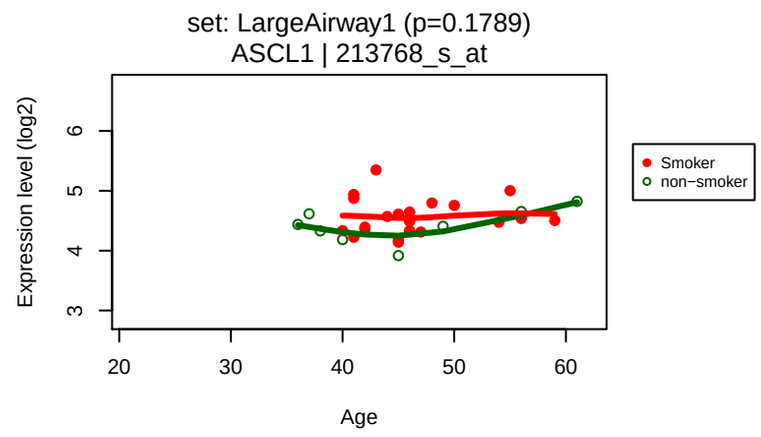

Gene expression of smokers and nonsmokers
gene: ASCL1 | 213768_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0568
p=0.2674
p=0.0089
p=0.1789
p=0.5818
p=0.4494
p=0.5506
p=0.0591
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

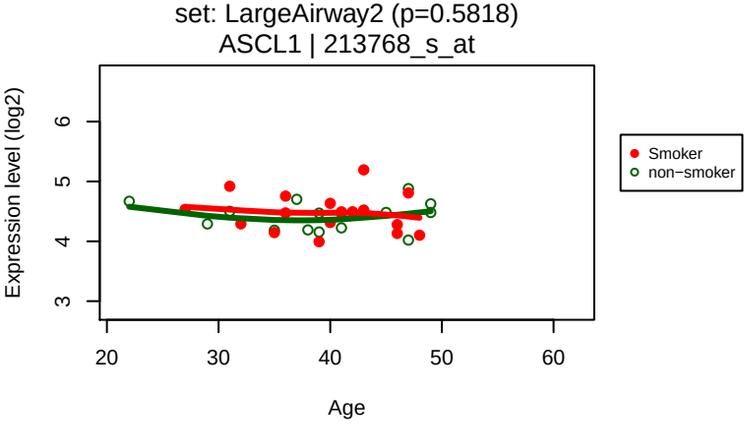

Gene expression ratio (smokers/nonsmokers)
gene: ASCL1 | 213768_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0568
p=0.2674
p=0.0089
p=0.1789
p=0.5818
p=0.4494
p=0.5506
p=0.0591
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

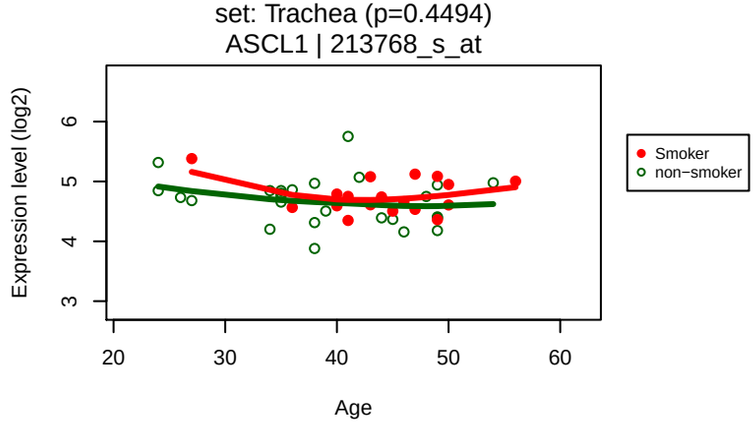

Gene expression ratio in Males
gene: ASCL1 | 213768_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4965
p=0.8886
p=0.1035
p=0.0656
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ASCL1 | 213768_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5875
p=0.2722
p=0.5928
p=0.4632
SmallAirway2
LargeAirway2
Trachea
Macrophage2

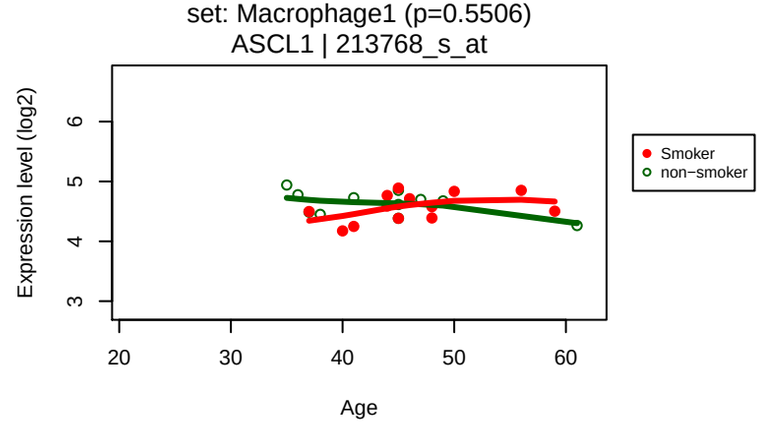

set: SmallAirway1 (p=0.0568)
ASCL1 | 213768_s_at
Expression level (log2)
Age
Smoker
non−smoker

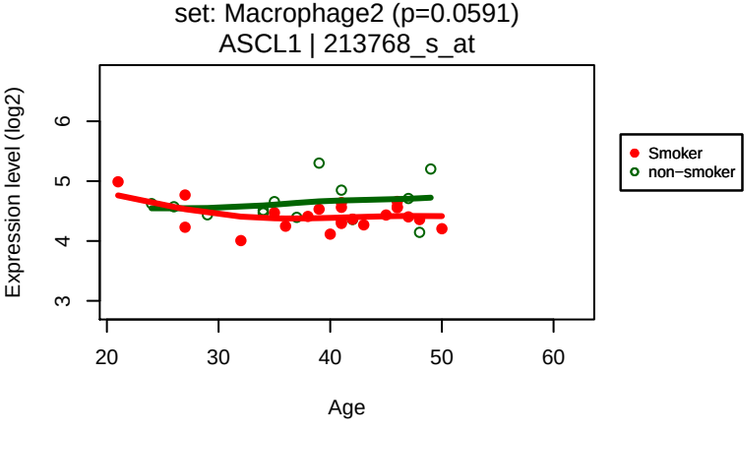

set: SmallAirway2 (p=0.2674)
ASCL1 | 213768_s_at
Expression level (log2)
Age
Smoker
non−smoker

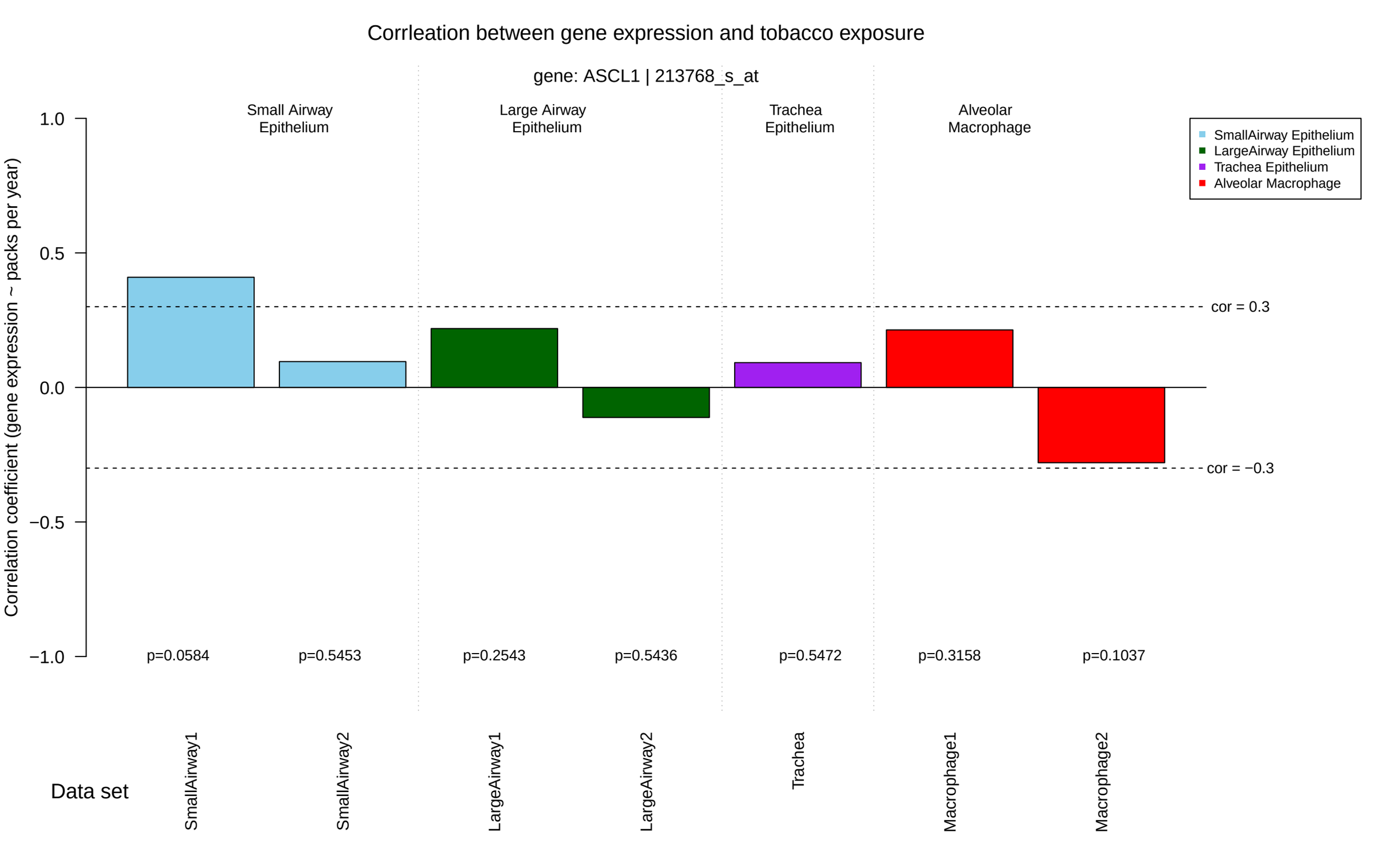

set: LargeAirway1 (p=0.1789)
ASCL1 | 213768_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5818)
ASCL1 | 213768_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.4494)
ASCL1 | 213768_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5506)
ASCL1 | 213768_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0591)
ASCL1 | 213768_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ASCL1 | 213768_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0584
p=0.5453
p=0.2543
p=0.5436
p=0.5472
p=0.3158
p=0.1037
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2